

Project Initialization and Planning Phase

Date	5 July 2026
Team ID	
Project Title	Smart Lender
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform loan approval using machine learning, boosting efficiency and accuracy. It tackles system inefficiencies, promising better operations, reduced risks, and happier customers. Key features include a machine learning-based credit model and real-time decision-making.

Project Overview	
Objective	The primary objective of Smart Lender is to revolutionize the traditional loan approval process by implementing an XGBoost-based machine learning model that evaluates applicant credibility using 11 key financial and personal features, ensuring faster, data-driven, and more accurate loan eligibility predictions.
Scope	The project covers end-to-end development — from data preprocessing and model training (Decision Tree, Random Forest, KNN, XGBoost) to deploying a Flask web application where applicants submit their details and receive an instant Approved or Rejected prediction with a detailed applicant summary and key decision factors.
Problem Statement	
Description	Addressing inaccuracies and inefficiencies in the current loan approval system adversely affects operational efficiency and customer satisfaction. Traditional loan processing is slow, biased, and lacks transparency — applicants are often rejected without understanding which factors (credit history, income, education) influenced the decision.
Impact	Solving these issues will result in improved operational efficiency, reduced risks, and an overall enhancement in the lending process, contributing to customer satisfaction and organizational success. Smart Lender reduces manual workload for bank officers and empowers applicants to make informed financial decisions.
Proposed Solution	
Approach	Employing machine learning techniques to analyze and predict creditworthiness, creating a dynamic and adaptable loan approval system. Four classifiers (Decision Tree, Random Forest, KNN, XGBoost) were trained and compared on the Kaggle loan dataset. XGBoost was selected as the best model with 81.1% test accuracy and deployed via a Flask web application.

Key Features	• Implementation of a machine learning-based credit assessment model — XGBoost trained on 11 features: Gender, Marital Status, Dependents, Education, Self-Employment, Applicant Income, Co-applicant Income, Loan Amount, Loan Term, Credit History, Property Area • Real-time decision-making for quicker loan approvals — prediction delivered instantly via POST /predict route in Flask • Continuous learning to adapt to evolving financial landscapes — model can be retrained anytime by running train_model.py with updated data, replacing model.pkl without any code changes • Transparent result page showing applicant summary table and key decision factors (Credit History, Income, Education, Property Area) • Graceful error handling and resubmission support via "Apply Again" and "Review & Resubmit" buttons
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Resource Requirements

Resource Type	Description	Specification / Allocation
Hardware		
Computing Resources	CPU/GPU specifications	Standard CPU (Intel i5 or above); T4 GPU for faster model training in Colab
Memory	RAM specifications	Minimum 8 GB RAM
Storage	Disk space for data, models, and logs	~50 MB for dataset + model.pkl; 1 TB SSD recommended for full development setup
Software		
Frameworks	Python web framework	Flask 3.0.3
ML Libraries	Machine learning & data processing	XGBoost 2.1.0, Scikit-learn 1.5.1, Pandas 2.2.2, NumPy 1.26.4, Matplotlib 3.9.0, Seaborn 0.13.2
Development Environment	IDE	Jupyter Notebook 7.2.1, PyCharm / VS Code
Data		
Dataset	Source, size, format	Kaggle Loan Prediction Dataset — 614 records, 13 columns, CSV format (dataset/loan.csv)